A

Mini Project Report

on

MySociety

Submitted in partial fulfillment of the requirements for the degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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This to certify that the Mini Project report on MySociety has been submitted by Krupa Devar (24107065), Palak Dey (24107098), Lavesh Dayme (24107125), Sandeep Devasani (24107053) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in Computer Science Engineering (Data Science), during the academic year 2025-2026 in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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CHAPTER 1

Introduction

In today's digital era, mobile applications have transformed the way everyday tasks are performed. However, many residential housing societies still rely on traditional and manual methods for managing their daily operations, such as maintaining physical registers for complaints, using notice boards for communication, and handling maintenance payments in cash or offline modes. These systems are inefficient, time-consuming, and prone to errors. The project *MySociety – Smart Housing Society Management App* aims to provide a centralized digital platform that simplifies and automates these operations. It enhances communication between residents and administrators while ensuring transparency, efficiency, and security in managing society-related activities.

The application introduces a modern solution where residents can interact with society services through their smartphones, and administrators can manage all operations efficiently from a single dashboard. This reduces dependency on manual work and improves the overall functioning of the society. The major contribution of this project is the integration of multiple features into a single application, including complaint handling, payment tracking, visitor management using QR codes, and real-time notice updates. Additionally, the system promotes a paperless environment, contributing to eco-friendly practices, and minimizes human errors by maintaining accurate digital records that can be easily accessed and updated in real time.

Furthermore, the application ensures better accountability among administrators by tracking actions and updates within the platform. The inclusion of notification systems keeps residents informed about important announcements, payment reminders, and complaint updates instantly, reducing communication gaps and enhancing community engagement. Designed with scalability in mind, the application can be implemented across both small and large housing societies with minimal customization. Its user-friendly interface allows individuals of all age groups to easily adapt to the system. Overall, by leveraging modern technologies, MySociety demonstrates how digital transformation can significantly improve the quality of life in residential communities and serves as a reliable, efficient, and future-ready solution for smart society management.

1.1 Purpose

The purpose of this document is to present the design, development, and implementation details of the *MySociety* mobile application. It provides a comprehensive understanding of the system architecture, features, technologies used, and the overall workflow of the application. The document also explains how different modules of the system interact with each other to deliver a seamless user experience. It highlights the importance of digital transformation in residential society management and how technology can simplify daily operations. The report serves as a complete reference for understanding both the technical and functional aspects of the project.

This document is intended for:

- Faculty members for academic evaluation
- Students for learning and reference
- Developers interested in similar mobile application projects
- Society administrators who can understand the benefits of digital society management

The report aims to clearly explain how the application solves real-world problems faced in housing societies and how modern technologies like Flutter and Firebase can be used to build scalable and reliable solutions.

1.2 Problem Statement

Managing residential societies using traditional methods leads to several inefficiencies and challenges. Residents often face difficulties in registering complaints and tracking their resolution status. Important notices frequently go unnoticed due to the absence of proper digital communication channels. Maintenance payments are difficult to track and may lack transparency, leading to disputes between residents and management. These manual processes are not only time-consuming but also prone to human errors, mismanagement, and delays in communication. There is a clear need for a unified digital solution that addresses all these challenges in an integrated and user-friendly manner. Such a system should provide real-time updates, secure data handling, and easy accessibility for both residents and administrators. It should reduce dependency on manual work and ensure transparency in all operations. A mobile-based platform can significantly improve communication, security, and efficiency within housing societies.

1.3 Objectives

The primary objectives of the MySociety application are as follows:

- To develop a user-friendly mobile application for housing society management
- To replace manual complaint systems with a digital complaint tracking system
- To enable real-time communication between residents and administrators through a digital notice board
- To maintain accurate and transparent records of maintenance payments
- To implement a secure visitor management system using QR code technology
- To provide a facility for booking society amenities such as a gym, clubhouse, or swimming pool
- To design a role-based system where different users (admin and residents) have controlled and appropriate access
- To reduce paperwork, human errors, and time consumption in day-to-day society operations
- To ensure data security and reliability using Firebase backend services

1.4 Scope

The scope of the MySociety application covers all major functionalities required for efficient housing society management. The system is designed to be scalable and adaptable for different types and sizes of residential societies. The application includes the following modules: Digital complaint management system, Notice board system for announcements and updates, Maintenance payment tracking and history, Visitor entry management using QR codes, Amenity booking system, and Admin dashboard for complete operational control. The system is accessible via Android mobile devices, making it convenient for users to access services anytime and anywhere.

The scope of the system is currently limited to software-based solutions and does not include integration with physical hardware systems such as biometric scanners or IoT devices. However, the architecture is designed in a way that allows future expansion with minimal changes. The application can later be enhanced by adding advanced features such as automated security systems, AI-based complaint classification, and predictive maintenance alerts. It also has the potential to integrate with cloud-based analytics tools for better decision-making. Another possible extension includes multi-platform support such as web and iOS applications. Overall, the system is designed with flexibility and scalability in mind, ensuring that it can evolve according to future technological advancements and user requirements.

CHAPTER 2

Proposed System

The proposed system, MySociety, is a mobile application developed using Flutter (Dart) for the frontend and Firebase as the backend. It is designed to modernize and simplify residential society management by replacing traditional manual processes with a digital platform. The system follows a role-based architecture where different users such as residents and administrators have access to specific functionalities based on their roles. This ensures better security, control, and efficiency in system operations.

Residents are the primary users of the application who interact with daily society services through the mobile app. They can easily access essential features such as complaints, payments, notices, and visitor management. Administrators are responsible for managing the overall functioning of the society through a centralized dashboard. They handle approvals, monitor activities, and ensure smooth operation of all services.

2.1 Features and Functionality

The application replaces traditional manual methods with a smart digital solution, enabling real-time communication and seamless data management. It ensures that all society-related tasks are performed quickly, efficiently, and securely through a mobile interface.

Resident Features

1. User Registration and Login : Residents can securely register and log in using Firebase Authentication. This ensures that only verified users can access the application. It also protects personal data through secure authentication mechanisms.

2. Complaint Management System : Users can raise complaints directly through the app and describe their issues in detail. They can also track the status of their complaints in real time until resolution. This improves transparency and response efficiency.

3. Digital Notice Board : Residents can view important notices posted by the admin instantly on their mobile devices. This ensures that all updates are received on time without relying on physical notice boards or manual communication.

4. Maintenance Payment Tracking : Users can view their pending dues and complete payment history anytime. This feature helps maintain financial transparency and allows residents to keep track of all transactions easily.

5. QR-Based Visitor Management : Residents can generate QR codes for their expected visitors through the app. Security personnel can scan these codes at the entrance, ensuring fast, secure, and verified entry into the society.

6. Amenity Booking System : Residents can book society facilities such as the gym, clubhouse, or swimming pool through the app. This reduces scheduling conflicts and removes the need for manual booking registers.

Admin Features

1. User Approval System : Admins can approve or reject new resident registrations based on verification. This ensures that only genuine and authorized users are allowed to access the system.

2. Notice Management : Admins can create, update, or delete notices as required. This enables quick and effective communication of important updates to all residents in real time.

3. Complaint Handling : Admins can view all complaints raised by residents along with their details. They can update and resolve issues efficiently, improving service quality and resident satisfaction.

4. Payment and Expense Tracking : Admins can record maintenance payments and monitor overall society expenses. This ensures proper financial management and improves transparency in accounting processes.

5. Amenity Booking Control : Admins can approve or reject booking requests for society amenities. : This helps in proper scheduling and ensures fair usage of shared facilities among residents.

CHAPTER 3

Project Outcomes

The MySociety application successfully transforms traditional housing society management into a modern, digital, and efficient system. It eliminates the need for manual paperwork, allowing residents to raise complaints, check notices, and access important information directly through their mobile devices. This significantly improves convenience and saves time. Communication between residents and administrators is also enhanced, as real-time notifications ensure that important updates are instantly delivered, reducing reliance on physical notice boards and manual methods.

The application also improves transparency and security within the society. Residents can easily track their maintenance dues and payment history, while administrators can efficiently manage financial records, reducing errors and building trust. The QR-based visitor management system ensures secure and controlled entry of visitors. Additionally, the app provides a centralized platform for handling complaints, bookings, notices, and payments, ultimately improving overall efficiency and enhancing the quality of living in residential societies.

Achieved Outcomes

Digital Transformation: The system successfully replaces manual processes with a fully digital platform, significantly reducing paperwork and improving overall operational efficiency.

Improved Communication: Notices and announcements are shared with all residents instantly through push updates, ensuring everyone stays well-informed at all times.

Efficient Complaint Handling: Residents can raise complaints digitally, and these can be tracked and resolved in a timely manner, improving satisfaction and trust in the management.

Secure Data Management: Firebase ensures safe and encrypted storage of all user data, significantly reducing the risk of data loss or unauthorized access.

Better Transparency: Payment and expense tracking is clear, accessible, and verifiable by both residents and administrators, which builds confidence and accountability within the society.

User Convenience: The mobile-based system allows residents and administrators to access all services anytime and anywhere, enhancing ease of use and accessibility.

CHAPTER 4

Software requirements

This chapter describes the tools, technologies, and system requirements used for developing the **MySociety application**. The application is designed using modern cross-platform and cloud-based technologies to ensure scalability, efficiency, and ease of maintenance. The frontend is developed using Flutter and Dart, providing a responsive and visually appealing user interface. Firebase is used as the backend for authentication, database management, and real-time data synchronization. Cloud Firestore is used to store structured data efficiently and support real-time updates.

Tools and Technologies

Frontend

Flutter 3.19 (Dart 3.3)
Flutter is used to build a responsive and cross-platform user interface for Android devices. It enables fast development using a single codebase while providing an expressive and attractive UI design. Its widget-based architecture ensures high performance and smooth user experience.

Backend

Firebase (Firebase SDK 10.x)
Firebase provides backend services such as authentication, real-time database, and cloud functions. It ensures seamless data synchronization across all users and devices, making the system highly responsive and reliable.

Database

Cloud Firestore
Cloud Firestore is a NoSQL database used to store all application data in a structured format. It supports real-time updates, offline data access, and high scalability, making it suitable for handling dynamic society management data.

Authentication

Firebase Authentication

Firebase Authentication provides secure login functionality using email and password. It handles user identity verification, session management, and security, ensuring safe access to the application.

Development Environment

- IDE: Android Studio Hedgehog (2023.1.1) / Visual Studio Code 1.87
- Operating System: Windows 10 / 11 (64-bit)
- Version Control: Git and GitHub

Minimum Device Requirements

- Operating System: Android 8.0 (Oreo) or higher
- RAM: Minimum 2 GB
- Internet Connection: Required for real-time data synchronization

Requirement Analysis

The system is designed to be lightweight, efficient, and scalable by utilizing cloud-based Firebase services. This reduces dependency on high-end hardware and ensures smooth performance even on low to mid-range Android devices. Real-time data synchronization enhances user experience by providing instant updates and responsiveness.

Additionally, cloud storage enables automatic backup and recovery of data, reducing maintenance efforts. The modular architecture of the system allows easy future enhancements without affecting existing functionality. Overall, the system ensures reliability, performance efficiency, and optimal resource usage. The application is developed with a focus on minimal system requirements so that it can be accessed by a wide range of users without performance issues. Since most processing is handled on the cloud, the client-side application remains lightweight and fast. This improves application stability and reduces device storage consumption. The use of Firebase services also eliminates the need for dedicated server management, making the system easier to deploy and maintain.

Furthermore, the system is designed to handle increasing numbers of users and data without performance degradation. Its scalable architecture ensures that additional features can be integrated in the future with minimal changes to the existing codebase. This makes the application suitable for real-world deployment in large residential societies with growing demands.

CHAPTER 5

Project Design

The design of the MySociety application follows a client-server architecture where the Flutter frontend communicates with the Firebase backend through the Firebase SDK. The Flutter application handles all user interactions and renders data, while Firebase manages data storage, authentication, and real-time synchronization. The database is structured using Firestore collections such as users, complaints, notices, payments, and visitors. The system workflow includes user registration, admin approval, login, and execution of various functionalities. The user interface is designed to be simple, intuitive, and easy to navigate. Security measures such as Firebase Authentication and role-based access control ensure safe and reliable operation.

System Architecture

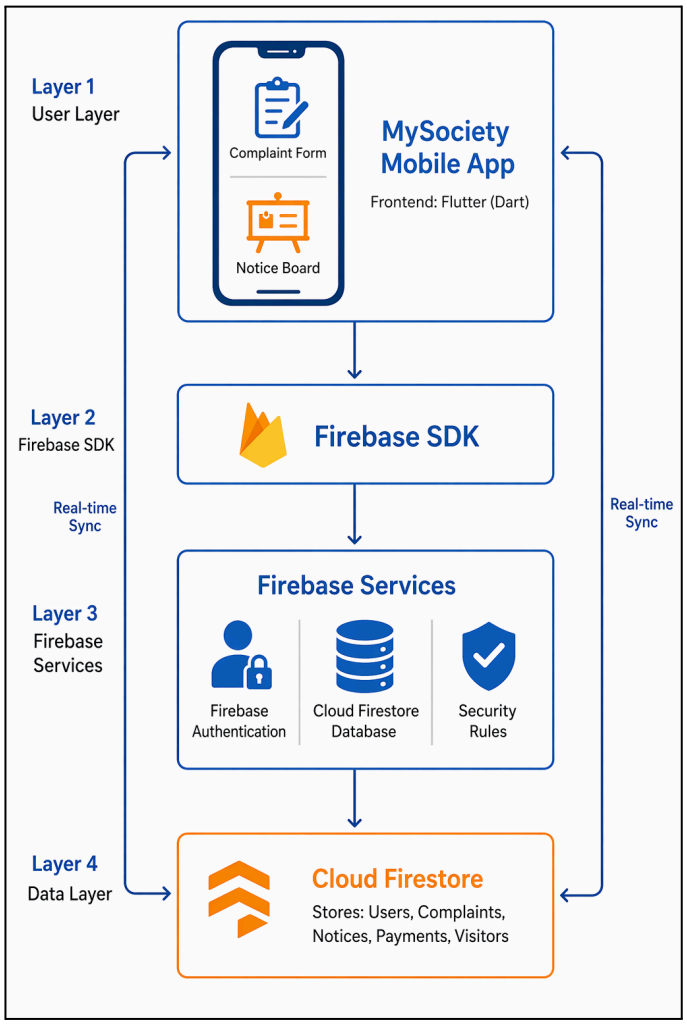


Figure 5.1: System Architecture

The system architecture of the MySociety application is organized into four layers. The first layer is the User Layer, which consists of the MySociety mobile application built using Flutter and Dart. This layer handles all user interactions including complaint submission and viewing of notices. The second layer is the Firebase SDK, which acts as the communication bridge between the mobile application and the Firebase backend services. The third layer contains the Firebase Services, which include Firebase Authentication for secure user login, Cloud Firestore Database for real-time data storage, and Security Rules for controlled data access. The fourth and final layer is the Data Layer, which uses Cloud Firestore to store all application data including users, complaints, notices, payments, and visitor records. Real-time synchronization is maintained between the mobile application and the data layer at all times.

Database Design

The application database is structured using the following Firestore collections:

Users Collection: Stores user details such as name, flat number, role (resident or admin), and contact information.

Complaints Collection: Stores complaint descriptions, categories, timestamps, and current resolution status.

Notices Collection: Stores announcements, posting dates, and content posted by the admin.

Payments Collection: Stores maintenance payment details including amount, due date, and payment status.

Visitors Collection: Stores visitor QR data, expected arrival time, and entry or exit records.

Security Design

Firebase Authentication is used for secure and verified user login. Role-based access control distinguishes between admin and resident permissions, ensuring each user can only access features relevant to their role. Firebase Security Rules are applied to protect all cloud data from unauthorized access. An admin approval system ensures that only valid and verified residents can register and use the application.

System Workflow

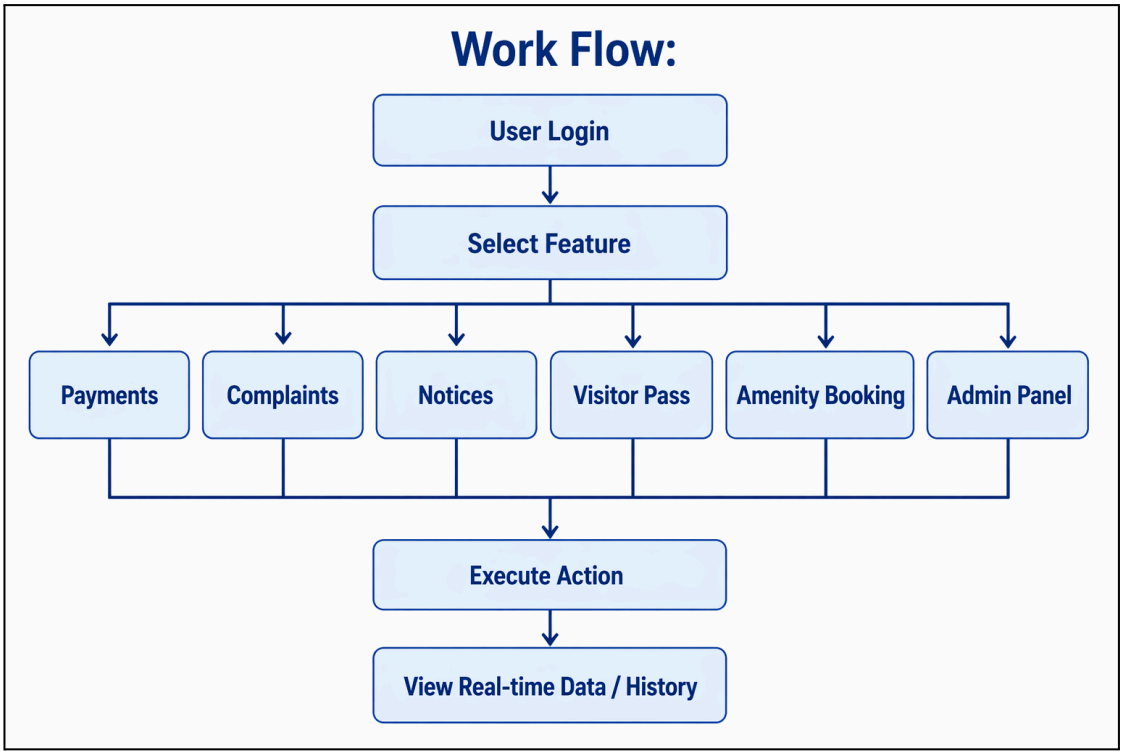


Figure 5.2: System Workflow

The system workflow of the MySociety application follows a simple and logical sequence of steps. The user first logs into the application using their registered credentials verified through Firebase Authentication. After login, the user is taken to the main screen where they can select any of the six available features: Payments, Complaints, Notices, Visitor Pass, Amenity Booking, or Admin Panel. Based on the feature selected, the user performs the required action such as raising a complaint, making a payment, or generating a visitor QR code. The action is then executed and all data is stored and synchronized in real-time through Cloud Firestore. Users can view updated data and history at any time through their dashboard.

User Interface Design

The user interface of the MySociety application is designed to be simple, clean, and user-friendly. It provides an intuitive dashboard-based layout that allows users of all age groups to easily understand and navigate through different features. The application ensures smooth navigation between screens with minimal steps, reducing complexity for all users. It is designed to be fully mobile-responsive, ensuring proper display and usability across different Android screen sizes and resolutions.

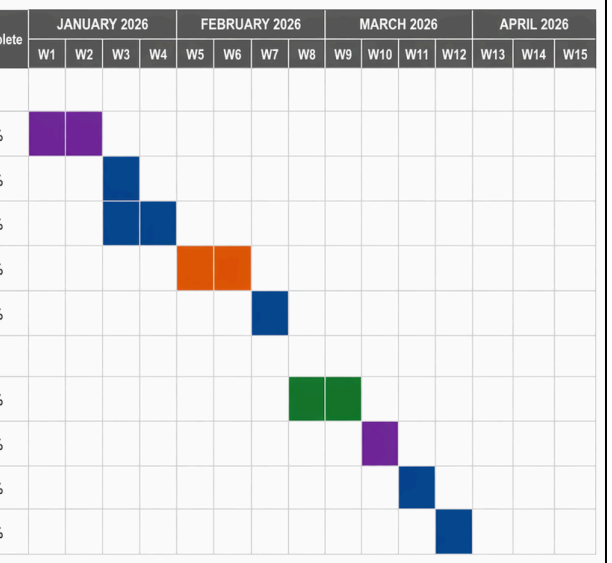
CHAPTER 6

Project Scheduling

The MySociety application was developed over four months, from January to April 2026, following a structured and phased approach. The project began with group formation, requirement analysis, and UI design using Flutter. This was followed by backend integration with Firebase and implementation of core features. System testing was then conducted to ensure proper functionality and reliability. Finally, project integration, documentation, and report preparation were completed. Proper scheduling ensured systematic and timely completion of the project.

MySociety – Project Scheduling Gantt Chart | January to April 2026

WBS No	Task Title	Task Owner	Start Date	End Date	Duration (Weeks)	% Complete
1	**Project Conception & Initiation**					
1.1	Group Formation & Topic Finalization	Palak Dey	5-Jan-26	16-Jan-26	2	100%
1.2	Identify Functionalities	Krupa Devar	16-Jan-26	20-Jan-26	1	100%
1.3	Discussion & Prototype	All	20-Jan-26	3-Feb-26	2	100%
1.4	GUI Design	Lavesh Dayme	3-Feb-26	17-Feb-26	2	100%
1.5	Presentation I	All	17-Feb-26	24-Feb-26	1	100%
2	**Project Design & Implementation**					
2.1	Database Design	Sandeep Devasani	24-Feb-26	10-Mar-26	2	100%
2.2	Database Connectivity	Palak Dey	10-Mar-26	17-Mar-26	1	100%
2.3	Integration & Report	Krupa Devar	17-Mar-26	24-Mar-26	1	100%
2.4	Presentation II	All	24-Mar-26	1-Apr-26	1	100%



Legend: Krupa Devar · Lavesh Dayme · Sandeep Devasani · Palak Dey

Figure 6.1: Project Scheduling (Gantt Chart)

The Gantt chart represents the complete project schedule of the MySociety application from 5 January 2026 to 1 April 2026, spanning 15 weeks and divided into two major phases. The first phase, Project Conception and Initiation, covers tasks such as Group Formation, Identifying Functionalities, Discussion and Prototype, GUI Design, and Presentation I. The second phase, Project Design and Implementation, includes Database Design, Database Connectivity, Integration and Report, and Presentation II. Each task is color-coded based on team member assignment — Dark Blue for Krupa Devar, Orange for Lavesh Dayme, Green for Sandeep Devasani, and Purple for Palak Dey. All tasks are marked 100% complete, confirming timely completion of the project.

CHAPTER 7

Results

The MySociety application was tested under various scenarios to ensure its functionality, reliability, and usability. The results obtained from testing confirm that the system performs effectively and meets all the intended objectives.

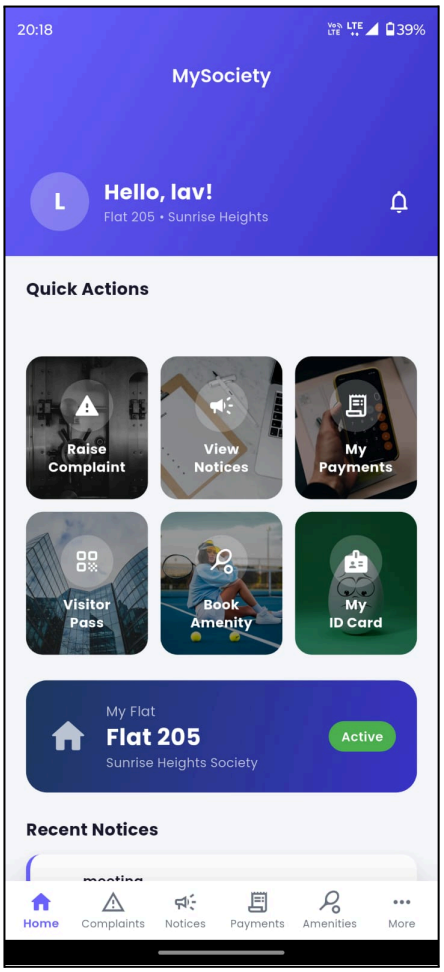


Figure 7.1: Home Dashboard Screen

Figure 7.1 shows the home dashboard of the MySociety application for residents. The screen displays a welcome message along with the user's flat number and society name. It provides quick access to key features such as raising complaints, viewing notices, checking payments, generating visitor passes, booking amenities, and accessing the digital ID card. The layout is simple and intuitive, allowing users to easily navigate between different functionalities. The dashboard also highlights recent notices, ensuring users stay updated with important information at all times.

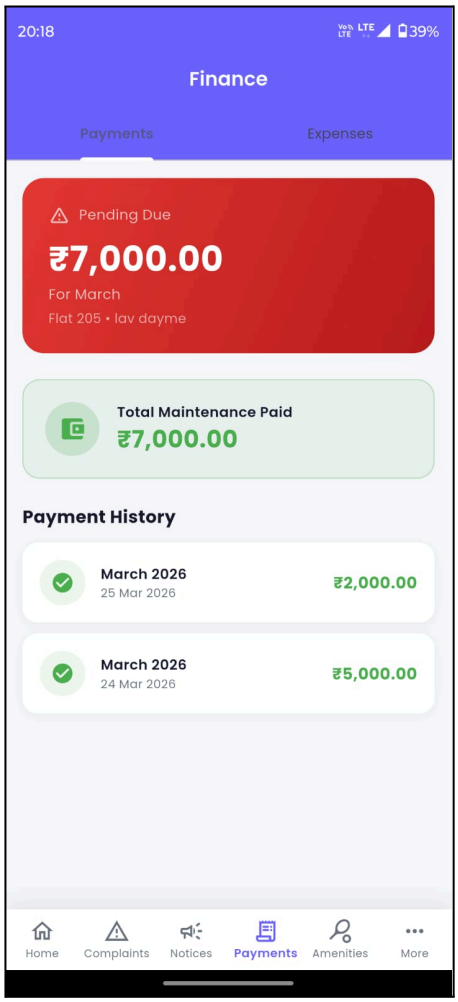


Figure 7.2: Finance and Payment Screen

Figure 7.2 represents the Finance module of the application, where users can view and manage their maintenance payments. The screen clearly displays pending dues along with the corresponding amount and month details. It also shows the total maintenance paid by the user for transparency. The Payment History section lists all previous transactions with their respective dates and amounts, helping users maintain clear and accurate financial records.

The Finance module is divided into two tabs — Payments and Expenses — allowing users to switch between different financial views easily. The pending due amount is highlighted in red, ensuring that users are immediately alerted about any outstanding payments. This visual indication helps reduce the chances of missed or delayed payments within the society. The module also records the exact date of each transaction, providing a reliable and detailed payment trail for future reference. Administrators can cross-verify these records from the admin panel to ensure consistency and accuracy in financial management. Overall, the Finance and Payment module promotes financial transparency and accountability, building trust between residents and the society administration.

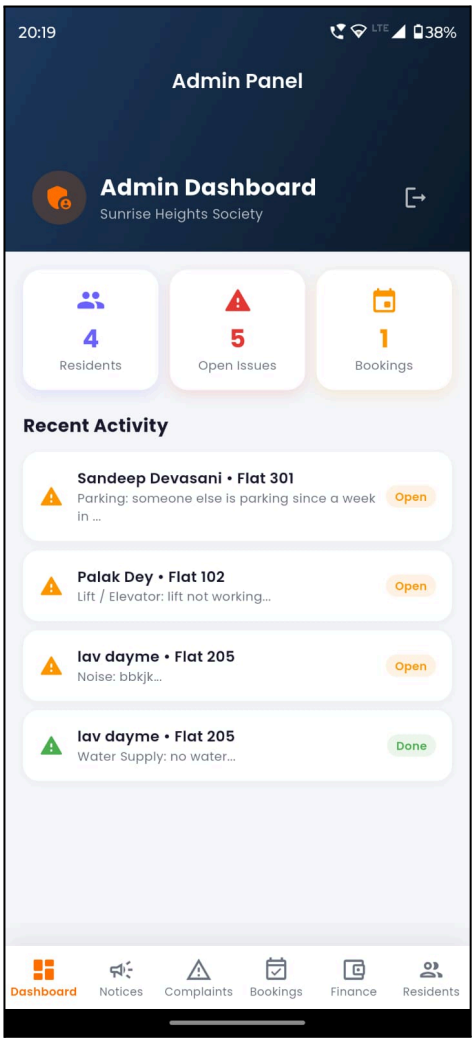


Figure 7.3: Admin Dashboard Screen

Figure 7.3 illustrates the Admin Dashboard, which provides administrators with a comprehensive overview of all society operations at a glance. It displays key statistics including the total number of registered residents, the number of open complaints, and active amenity bookings. The Recent Activity section shows all ongoing and resolved complaints along with their current status, allowing the admin to efficiently monitor and manage all issues from a single screen. Each complaint entry displays the resident's name, flat number, complaint category, and current resolution status, making it easier for the admin to prioritize and act on urgent issues. The dashboard also provides quick navigation to other admin modules such as Notices, Complaints, Bookings, Finance, and Residents through the bottom navigation bar. This centralized control panel eliminates the need for the admin to switch between multiple screens to get an overview of society activities. The clean and minimal design of the dashboard ensures that even non-technical administrators can use the system with ease and confidence. Overall, the Admin Dashboard serves as the command center of the MySociety application, enabling efficient, transparent, and real-time management of all housing society operations.

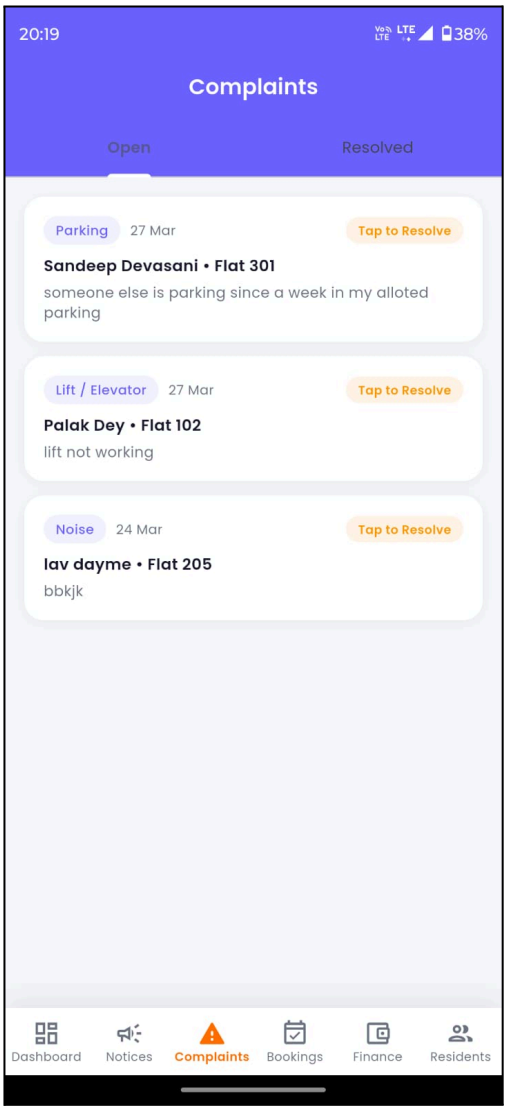


Figure 7.4: Complaint Management Screen

Figure 7.4 shows the Complaint Management module of the application. The screen lists all complaints raised by residents along with relevant details such as complaint category, date of submission, resident name, flat number, and a brief description. Each complaint displays a "Tap to Resolve" button, making it straightforward for the admin to take action and update the complaint status. This feature ensures timely resolution of issues and improves communication and trust between residents and the society management. The complaints are categorized under different types such as Parking, Lift/Elevator, Noise, and Water Supply, allowing the admin to quickly identify and prioritize critical issues. Resolved complaints are moved to a separate tab, keeping the active complaint list clean and manageable for the admin. Overall, the Complaint Management module plays a vital role in maintaining a smooth and efficient society operation by ensuring that no resident concern goes unaddressed.

Test Cases

The following test cases were executed to validate the core functionalities of the MySociety application:

Test No.	Module	Test Description	Expected Result	Actual Result	Status
TC-01	Authentication	Resident login with valid credentials	User is redirected to Home Dashboard	Redirected to Home Dashboard	Pass
TC-02	Authentication	Login with invalid credentials	Error message displayed	Error message displayed	Pass
TC-03	Complaint	Resident raises a new complaint	Complaint saved and visible to admin	Complaint saved and reflected in admin panel	Pass
TC-04	Admin	Admin resolves a complaint	Complaint status updated to Resolved	Status updated to Resolved	Pass
TC-05	Payments	Resident views payment history	Payment history listed with dates and amounts	Payment history displayed correctly	Pass
TC-06	Visitor	Resident generates a QR visitor pass	QR code generated successfully	QR code generated and displayed	Pass
TC-07	Notices	Admin posts a new notice	Notice visible to all residents immediately	Notice appeared on resident dashboard	Pass
TC-08	Amenity	Resident books an amenity	Booking request submitted to admin	Booking request created successfully	Pass

CHAPTER 8

Future Scope

While the MySociety application successfully addresses the core requirements of housing society management, there are several areas where the system can be further enhanced in future versions. The current application supports Android devices only. Future development can extend the application to iOS using Flutter's cross-platform capabilities, making it accessible to a wider user base. A dedicated online payment gateway such as Razorpay or PayTM can also be integrated into the payment module, allowing residents to pay their maintenance dues directly through the application without requiring manual offline payments. Additionally, push notification support can be enhanced to alert residents immediately when a complaint is resolved, a new notice is posted, or a payment is due, which would further improve real-time communication within the society.

The system can be integrated with IoT-based physical hardware devices such as biometric scanners, smart gate controllers, and CCTV camera feeds to further enhance security and automate visitor entry without manual QR scanning. A detailed analytics dashboard for administrators can be developed to provide insights such as monthly complaint trends, payment collection rates, and amenity usage statistics, enabling data-driven decision-making by society management. Furthermore, an AI-based complaint categorization system can be introduced that automatically assigns complaints to the relevant department based on keywords, reducing manual intervention and improving resolution time. The application can also incorporate a community chat feature that allows residents of the same society to communicate, share updates, and organize events through a secure in-app messaging platform.

The system can be expanded to support multiple societies under a single administrative platform, making it suitable for larger real estate groups managing several residential complexes simultaneously. A multilingual interface can be introduced to make the application more accessible to residents who are more comfortable in regional languages such as Hindi, Marathi, or Bengali. The application can also be enhanced with a document management module that allows residents to digitally store and access important documents such as society bylaws, maintenance receipts, and meeting minutes. With these future enhancements, MySociety has the potential to evolve into a comprehensive smart society management platform that caters to the diverse and growing needs of modern residential communities.

CHAPTER 9

Conclusion

The MySociety application is a comprehensive and practical solution designed to modernize residential society management. Through this project, the team has successfully demonstrated how mobile technology and cloud-based backend services can be used to address real-world problems effectively and efficiently. The application replaces traditional manual processes with a clean digital platform that is efficient, transparent, and easy to use for all age groups. It enhances communication between residents and administrators, improves security through QR-based visitor management, and simplifies financial management through transparent payment tracking, making it genuinely beneficial for both residents and administrators.

One of the key strengths of the project is its user-friendly interface, which ensures that users of all technical backgrounds can easily navigate and use the application. The use of Firebase enables real-time data synchronization, making the system highly responsive, reliable, and available at all times. The project also highlights the importance of role-based access control in ensuring that users only interact with features relevant to their role, improving both usability and system security simultaneously. The integration of multiple society management features into a single unified platform eliminates the need for separate tools, reducing complexity for both residents and administrators. Furthermore, the use of Flutter ensures that the application delivers a consistent and smooth experience across different mobile devices and screen sizes.

In conclusion, MySociety is a scalable, practical, and extensible solution that can be readily implemented in real-world housing societies of any size. It not only improves operational efficiency and reduces administrative workload but also enhances the overall living experience and quality of life for residents. As technology continues to evolve, the application can be further extended with features such as IoT-based gate automation, biometric access control, and AI-powered complaint categorization. This project stands as a strong example of how student-driven development can produce real, impactful, and deployment-ready software solutions. The team is confident that MySociety has the potential to bring meaningful digital transformation to residential communities across the country.

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